Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling

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Code, results, and reproducibility materials: `https://github.com/lsalcedo100/FusionFlux`
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Abstract

Confinement scaling laws set the size a next-step tokamak has to reach. On the International Tokamak Physics Activity (ITPA) Global H-mode Confinement Database (HDB5, standard set STD5; 6228 slices, 18 device/wall labels), under cross-validation grouped by discharge, a random forest reduces log error by 29% against a power law refitted within the same folds and 36% against the published IPB98(y,2) reference. That margin does not survive the question a scaling law exists to answer. Under leave-one-label-out the ordering reverses: the forest is worse than that power law on all 13 held-out labels and all 11 physical devices. It survives nested tuning, threshold and fold-assignment sweeps and dropping the spherical tokamaks; equal label weighting leaves it at 12 of 13. The forest's error tracks the held-out label's Mahalanobis distance from the training distribution ($\rho = +0.85$) where the power law's does not, and it cannot predict outside its training range. At a cut reproducing ITER's 1.82× jump in major radius the trees score closer to a constant predictor than to the power law, and 90% split-conformal intervals fall to 3% and 0% coverage at unchanged width. Two repairs help, both supplying long-range structure rather than capacity: Connor–Taylor constraints on the exponents, whose Kadomtsev rung is free in sample and beats the unconstrained fit at every well-powered size cut, and a Gaussian process whose linear component keeps trending. Across the families tested, extrapolation failure tracks long-range saturation rather than flexibility. For ITER the IPB98(y,2) and forest predictions differ by 8.3×, the forest's being its training ceiling rather than an estimate.

Keywords: tokamak energy confinement; confinement scaling; machine learning; extrapolation; cross-validation; dimensional similarity; uncertainty quantification; ITER

1 Introduction

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, which makes it a

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principal input to how large a next-step device has to be. Empirical scaling remains a principal tool for projecting it to a device that does not exist yet [1]: a power law in the engineering parameters,

$$\tau_E = C\, I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \tag{1}$$

fitted by least squares across discharges from the multi-machine confinement database. In log space Eq. (1) is exactly a linear model, so fitting one is ordinary least squares on a log design matrix. The reference fit is IPB98(y,2) [1], a regression over the ITPA multi-machine database whose exponents were used to size ITER; that database is still maintained and republished [2].

What such a law is used for is an extrapolation, and a large one. ITER's major radius is 1.82 times the largest machine in the database, and SPARC [3] is designed for a toroidal field 2.1 times the largest recorded in it. The law is asked for a number at a point no row occupies, and that is the regime in which a fitted regression carries no guarantee at all.

This paper reports two things about that regime, and both are worth separating from the expectation that accuracy simply degrades outside the data. The first is that the *ranking* of candidate models reverses. Under cross-validation grouped by discharge a random forest beats a power law refitted in the same folds by 29%; held out one whole device at a time, it is worse than that same power law on every device scored. A practitioner following the conventional procedure does not merely get an optimistic score for the right model, they get the wrong model. The second is that the calibrated intervals fail *without widening*. Split-conformal intervals set to 90% cover 3% and 0% of held-out rows at a size cut matched to ITER's, at essentially unchanged width: they are confidently wrong rather than usefully vague, which is the failure a design study cannot detect by inspecting the interval it is handed. Degradation is a matter of degree and can be absorbed by a margin. Neither of these can.

A third observation follows from the same table and points at what to do about it. On a held-out label the published IPB98(y,2) law scores 0.188 against 0.214 for the power law refitted here on the same rows and features, and 0.194 against 0.278 at the size cut. A law fitted in 1998 to a predecessor database transfers better than an unconstrained refit of the same functional form on more data. That is not an accident of which rows each saw. IPB98(y,2) was fitted under an enforced high-β Kadomtsev constraint with the field exponent held at 0.15 [1, 2], which is exactly the long-range structure Sec. 8 finds to be worth having: imposing the same family of constraints on a refit here recovers most of the gap without costing anything in sample. The constructive half of this paper follows from that, and the ranking inversion is what makes it necessary rather than optional.

Replacing Eq. (1) with a flexible regressor is an obvious move, and machine learning has been productive elsewhere in this field: disruption prediction [4], transfer of a disruption predictor to a device that supplied almost none of its training data [5], and real-time magnetic control [6]. Confinement scaling itself has been revisited with symbolic regression, which abandons the power-law form rather than fitting it [7], and with a neural residual learned on top of a frozen power law [8]. Measured conventionally the flexible models win. On the rows used here the random forest scores 0.128 in log-RMSE against 0.181 for a power law refitted inside the same folds on the same rows and features, a margin of 29%, and against 0.199 for the published IPB98(y,2) law, a margin of 36%.

The two comparators are not interchangeable and both are quoted throughout. IPB98(y,2) is a historical reference: it was derived in 1998 from version 2.8 of this database's predecessor (DB2.8) rather than from the DB5.2.3 revision analysed here [2]. That predecessor overlaps the present database substantially in devices and operating regimes, so for most folds below the machine being held out contributed to the data IPB98(y,2)'s exponents were fitted on. It is therefore not a fold-specific comparator that can be described as having been fitted without the held-out rows, and no claim in this paper rests on treating it as one. The power law refitted inside each fold is the like-for-like baseline, and the reversal below is stated against it. Every number in this paper is scored on the nine engineering features; the ten-column matrix of Sec. 3, which adds IPB98(y,2)'s own prediction to them, is reported there as a control and nowhere else.

Related work on this database. Four recent studies apply machine learning to the same ITPA data and are the closest prior work. Nam and Seo [9] take cross-machine transfer as their subject: they align features to correct the covariate shift between devices and use deep ensembles for uncertainty, validating on conditions the model has not seen, including JET-ILW and ITER. Gao *et al.* [10] apply variational and sampling-based Bayesian neural networks to STD5 together with individual JET, DIII-D and ASDEX Upgrade data, reporting improved accuracy alongside posterior uncertainty. Xiong *et al.* [11] tune random forests and two neural architectures by Bayesian optimisation on STD5 and report R^2 up to 0.990, with a feature-importance analysis separating spherical from conventional tokamaks. Wang *et al.* [8] freeze a power law and learn its residual, reporting improved stability under shifts defined by parameter ranges.

Nam and Seo [9] in particular already validate on unseen conditions and report that cross-machine transfer is harder. That is the claim this literature generally supports, and it is not the claim made above: accuracy degrading out of distribution is a matter of degree, and a margin can absorb it, where a reversed ranking and an interval that misses without widening cannot.

None of these works is the target of a criticism here; what any of them reports is what its split measures. Where a score on this database is computed without holding out a whole device, it describes discharges from machines the model has already seen. We do not propose a better predictor. We ask whether the validation that selects one answers the question a scaling law exists for, measure what the honest answer costs, identify the property of a model that decides it, and test the repair that physics rather than flexibility supplies.

That measurement does not directly address the deployment problem of interest, for a reason about the data rather than about the models. Cross-validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so a held-out row has hundreds of near neighbours recorded on the same device at nearby settings. What the score quantifies is how much of JET is predictable from the rest of JET. That a validation split should match the generalisation being claimed, and that grouped observations flatter a model when it does not, is well understood in fields whose data arrives in clusters [12]; it is not what a commonly reported comparison on this database measures. The quantity a scaling law is built to supply is a prediction for a machine that does not yet exist.

This paper measures what the deployment-matched split costs. We escalate what is held out in four steps, a discharge, then a database label, then a physical device, then an entire size range, and score the same models under each. Under the second step the ranking of the three primary fitted models is the exact reverse of their ranking under grouped cross-validation, and the best cross-validated model loses to a log-linear power law on all 13 held-out database labels and all 11 physical devices. Three measurements say why, and they cover extrapolation distance, the training-target bound of a random forest, and what polynomial flexibility costs in the tail. At the fourth step, which reproduces ITER's extrapolation factor inside the database, conformal intervals calibrated to 90% cover 3% of rows without becoming any wider. Two repairs then supply the long-range structure the diagnosis implicates: the Connor–Taylor dimensional constraints imposed on the exponents, and a Gaussian-process experiment that separates flexibility from long-range saturation and corrects the diagnosis the earlier sections leave open. The result is replicated on rows the standard analysis set does not contain, in a second confinement regime, and, in Secs. S4 and S5 of the Supplementary Material, outside fusion. Section 3 begins with a property of the design matrix itself: it is rank deficient by two, and one of the two dependencies is that a published scaling law added as a model feature contributes nothing a log-linear fit can use.

2 Data and methods

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [2], obtained from the Open Science Framework (OSF). The delivered file contains 6228 rows from 4471 discharges across 18 database device/wall labels, which correspond to 16 physical devices once the two JET entries and the two ASDEX Upgrade entries are each collapsed. The cleaning rule below requires finiteness and positivity in every used column; on this file it removes no rows, so all 6228 delivered rows are analysed. The target is the thermal confinement time `TAUTH`. No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms. Its exponents were fitted on a predecessor of this database that overlaps it in devices and regimes, so for most folds it carries information about the held-out label; retaining it would import that overlap into every fold. The model set is a ridge regression on the log features (i.e. a fitted power law), a random forest [13] (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. IPB98(y,2) evaluated analytically is reported as a historical reference and is marked as such throughout: it is not refitted within folds, and its scores are not part of the ranking claim.

What is and is not tuned. Every model is fitted and scored in $\log \tau$: the target is $\log \tau$ and the reported error is the root-mean-square residual in the same variable. Exponentiation to τ enters only where a quantity is quoted in seconds, such as the mean absolute error and the device forecasts of Sec. 11. None of the models in Table 1 is hyperparameter-tuned, and "library defaults" is not a specification, so the consequential settings of every model, and of the polynomial and Gaussian-process rungs, are written out in full in Sec. S7 of the Supplementary Material rather than left to a default a later release may change.

Every estimator carries the seed 42, and the forest is fitted in parallel but predicts single-threaded so that reruns are bit-identical. Standardisation sits inside the pipeline, so it is refitted on the training rows of every fold and never sees a held-out row. No within-model hyperparameter search is performed on any reported cross-validation score, so nested tuning is not required for those numbers, and Sec. 4.2 reports what a nested search does when one is run anyway. Comparing several prespecified model families is still a selection of a kind, and two later choices go further: the constraint rung of Sec. 8 and the damping of Sec. 6 were both picked after seeing held-out scores, and both are flagged where they appear. The Gaussian processes of Sec. 12 do estimate kernel hyperparameters, by marginal likelihood on training rows alone.

Splits. (i) Grouped cross-validation by discharge: `GroupKFold` with five folds, grouped on the discharge identifier, so every slice from one discharge falls in the same fold. It is constructed with `shuffle=False`, its default, so the partition is deterministic and no seed enters it; Sec. 4.2 reports the same comparison over shuffled partitions. (ii) Holding out a whole unit, in two versions that are named separately throughout because the 18 labels are not 18 physical devices. *Leave-one-label-out* (LOLO in the tables) holds out each of the 13 database device/wall labels with at least 30 rows in turn, training on every other row in the database. *Leave-one-device-out* holds out each of the 11 physical devices those labels reduce to once JET and JET-ILW are merged, as are the two ASDEX Upgrade entries. The distinction matters here more than it usually would: holding out JET-ILW while JET stays in training does not hold out an unseen device, and this paper's own argument is that the held-out unit has to match the deployment being claimed. Leave-one-label-out is the default reported column because it is the finer partition and the one a practitioner would reach for first; every claim that turns on a device being genuinely unseen is stated against leave-one-device-out and labelled as such. The 30-row threshold governs only which labels are scored, not which are trained on, so the five labels below it (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them) stay in every training fold; each fold therefore trains on 17 labels. It was fixed before any held-out score was computed, as the smallest round number giving a per-label log-RMSE over enough rows to be worth reporting, and it is not a tuned quantity: Sec. 4.2 sweeps it over 10, 20, 30 and 50 rows and reports what the win count does, and also reports what happens when the five sub-threshold labels are dropped from training as well. (iii) A size-ordered cut: labels are ordered by median major radius, the model is trained below a cut and scored on every label above it, and the reported cut is the one whose size ratio is closest in log to the ratio between this database and ITER. That rung has a ratio of 1.823, trains on 14 database labels and scores 2730 held-out rows, and is called the ITER-size-matched cut throughout. The match is to ITER's major-radius ratio and to nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient.

The score is the root-mean-square error in $\log \tau$, $\mathrm{RMSE}_{\log \tau} = \sqrt{\overline{(\log \hat{\tau} - \log \tau)^2}}$, written log-RMSE here and in the figures. It is the plain log residual, not the $\log(1 + y)$ quantity usually called RMSLE; confinement times are strictly positive, so no offset is needed, and the name is avoided here to keep the two distinct. Intervals are percentile bootstraps resampling whole *units*, since the claim concerns behaviour on an unseen unit: 2000 resamples drawn with replacement over the 13 labels, seeded at 20240617, reported as the 2.5th and 97.5th percentiles. Paired intervals resample the per-label difference rather than the two error levels, which removes the shared unit difficulty.

What the inferential statistics here do and do not claim. Stated once, and referred back to rather than repeated. Every fold in a leave-one-unit-out arm trains on all the other units, so any two folds share almost all of their rows. The units they score are therefore not independent experiments, and nothing below treats them as such. Three consequences follow and hold throughout. The bootstrap intervals are summaries of spread over the held-out units, not confidence intervals under independent replication. The permutation and sign-test p-values, including the exact sign test under what is called the independent-unit-sign null in Sec. 4.2, are compact summaries of how lopsided a win count is, not evidence from n replicates. And the conformal coverages are empirical row-level rates rather than guaranteed ones, for the separate reason given in Sec. 7. What carries the claims in this paper is the descriptive win counts and the size of the gaps, which need none of that machinery; the statistics describe their spread.

Preprocessing. Rows are dropped only for non-finiteness or non-positivity in a used column, which on this file removes none of the 6228 delivered rows; no outlier rule, no winsorising and no reweighting is applied. Features are natural logs of the raw engineering columns, taken after cleaning. Minor radius is reconstructed as $a = \epsilon R$ at this stage, which is the origin of one of the two exact collinearities of Sec. 3. Wall variants (JET-C and JET-ILW, and the two ASDEX Upgrade entries) are separate labels in the database and are kept separate except where the text says devices are collapsed to 11.

Software. Python 3.12.14, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.18.1 and pandas 2.3.3. The version of every library is pinned in the repository's `constraints.txt`. Any setting left at its library default therefore resolves to those versions' default; the consequential settings are written out above rather than referenced, because a default may differ in a later release.

3 The design matrix, and what a refit can identify

Three column counts appear in this paper and they are not the same set. The models see nine log engineering features. Adding the analytic IPB98 prediction to those nine gives the ten-column matrix examined in this paragraph. The refit of Eq. (1) in Sec. 3.1 drops both redundant columns and uses nine, eight independent variables plus an intercept, which is why its condition number is a healthy 10.7.

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14}. Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16}: minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prediction is a fixed log-linear combination of the other eight features. The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. The deficiency raises no error, because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

Neither the redundant column nor the estimator is load-bearing. Dropping `log_a_m` and refitting on eight independent columns leaves the log-linear scores unchanged to four decimals under all three splits, moves the tree ensembles by at most 0.035, and changes no ordering anywhere. Ordinary least squares on those same eight columns, carrying no penalty at all, reproduces the reported ridge to six decimals under cross-validation and to 0.0012 at the size cut, so the ridge here is a numerically stable spelling of the least-squares power law rather than a regularised alternative to it. The models keep the redundant nine-column representation because minor radius is a parameter a practitioner would naturally supply, and promoting the cleaner eight-column variant on the strength of its scores would be exactly the selection Sec. 8 flags in its own result.

3.1 Refitted exponents and a weakly identified direction

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9, condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. The disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks, and a refit on a different population is a different regression. It is not a solver artefact either; independent implementations by the normal equations, by QR and by the SVD pseudoinverse agree to 7.6×10^{-13}.

Where the two differ matters more than by how much. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix's variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, and it is weak for a structural reason: tokamaks are not designed to vary the two independently, so the existing multi-machine design leaves the direction weakly identified. More shots at the operating points these machines already run would not separate it; shots deliberately placed to break the correlation could.

Why the size dependence in this database is weaker than IPB98(y,2)'s is itself an active question, and Hall *et al.* [14] attack it directly on DB5.2.3, treating the multicollinearity among the predictors explicitly and searching for the smallest subset of points that most reduces the fitted major-radius exponent. A continuation [15] attributes the weak size dependence predominantly to a subset of discharges localised in dimensionless space, distinguished by normalised pressure, collisionality, normalised gyroradius and safety factor rather than by any single device or by metallic walls, and reports that excluding it recovers a major-radius dependence broadly consistent with IPB98(y,2). That is a different question from the one asked here. They ask where the changing size dependence comes from; this paper takes the fitted exponents as given and asks how the choice of validation split changes which model is preferred. The two bear on each other: if the size direction is partly an artefact of a localised subset, then the size-ordered cut of Sec. 5 measures transfer across a boundary those authors would draw differently. That is a caveat on how much weight that one axis can carry, and not on the ranking reversal, which is measured per unit rather than per size.

4 The ranking inversion

Table 1 is the central result. Among the three primary models, a power law, a random forest and a gradient booster, the order under one split is the exact reverse of the order under the other. The random forest is the best of the three primary models under cross-validation and the worst of the three on a label it has not seen. It is not the best model in the paper: the Gaussian process of Sec. 12 scores 0.112. Both columns compare the same model specifications and the same features, but they differ in scoring population and in aggregation as well as in held-out unit; Sec. 4.2 matches those choices and shows the reversal survives. Against the fold-fitted power law the forest is 29% better in the CV column and 117% worse in the LOLO column. The analytic IPB98(y,2) row is the published reference: its exponents come from DB2.8, a predecessor of this database that overlaps it heavily in devices and regimes [2], so it is reported as a historical reference rather than as a model refitted within these folds. The two polynomial rungs are controls,

Table 1: The headline comparison under the conventional reporting choices: the same models and the same nine engineering features, scored one way per split (LOLO is leave-one-label-out over the 13 eligible labels). The two columns differ in more than the held-out unit, since they also differ in row population and in how per-row scores are aggregated, and the 13 labels are not 13 independent devices. Sec. 4.2 matches the population and the aggregation and repeats the comparison over physical devices; the inversion survives all of it. The rank columns cover only the six models above the rule, all of which are refitted inside every fold. IPB98(y,2) is set below the rule and carries no rank: it is not refitted within folds and is not part of the ranking claim.

Model	CV	LOLO	Ratio	CV rank	LOLO rank
random forest	0.128	0.465	3.6×	1	4
hist gradient boosting	0.130	0.359	2.8×	2	3
ridge, log-cubic	0.148	0.849	5.7×	3	5
ridge, log-quadratic	0.158	0.300	1.9×	4	2
ridge, log-linear	0.181	0.214	1.2×	5	1
mean baseline	0.869	0.994	1.1×	6	6
Historical reference, not fold-fitted					
IPB98(y,2), analytic	0.199	0.188	1.0×	n/a	n/a

scored in the same table but outside the ranking claim, and including them breaks the inversion rather than extending it: the log-cubic rung is third of the five fitted models under cross-validation and last of them on a held-out label.

Because the labels differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-label difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** database labels, mean gap +0.251, 95% interval $[+0.157, +0.342]$. The gradient booster is worse on 12 of 13. Figure 1 shows the reversal together with the three extrapolation diagnostics of Sec. 4.1. Section 4.2 repeats this over the 11 physical devices, where the labels are no longer two eras of one machine counted twice, and the gap widens to +0.315 with an interval of $[+0.221, +0.408]$.

4.1 Three diagnostics of the failure

Distance. Ranking each model's per-label error against the Mahalanobis distance of that label's mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and +0.54 for the gradient booster, against $\rho = -0.06$ for the log-linear power law. The random forest's error rises strongly with extrapolation distance and the gradient booster's trends upward more weakly. The power law's does not: its error is uncorrelated with how far the label sits from anything it saw.

Thirteen points is few enough that each of those numbers needs an uncertainty. Permuting the distances 20000 times gives $p = 0.0007$ for the forest, $p = 0.060$ for the booster and $p = 0.85$ for the power law. That permutation treats the 13 database labels as exchangeable, which they are not exactly: two physical devices contribute two wall-era labels each, so these p-values are read as Sec. 2 says: summaries of how lopsided a count is. Recomputed over the 11 physical devices, which removes the duplicated wall-era labels, the forest's correlation weakens slightly to +0.77 ($p = 0.005$) and the power law's is unchanged at −0.06 ($p = 0.85$), so the contrast the section rests on survives the collapse. That permutation p-value is read the same way. The booster's falls from +0.54 to +0.38

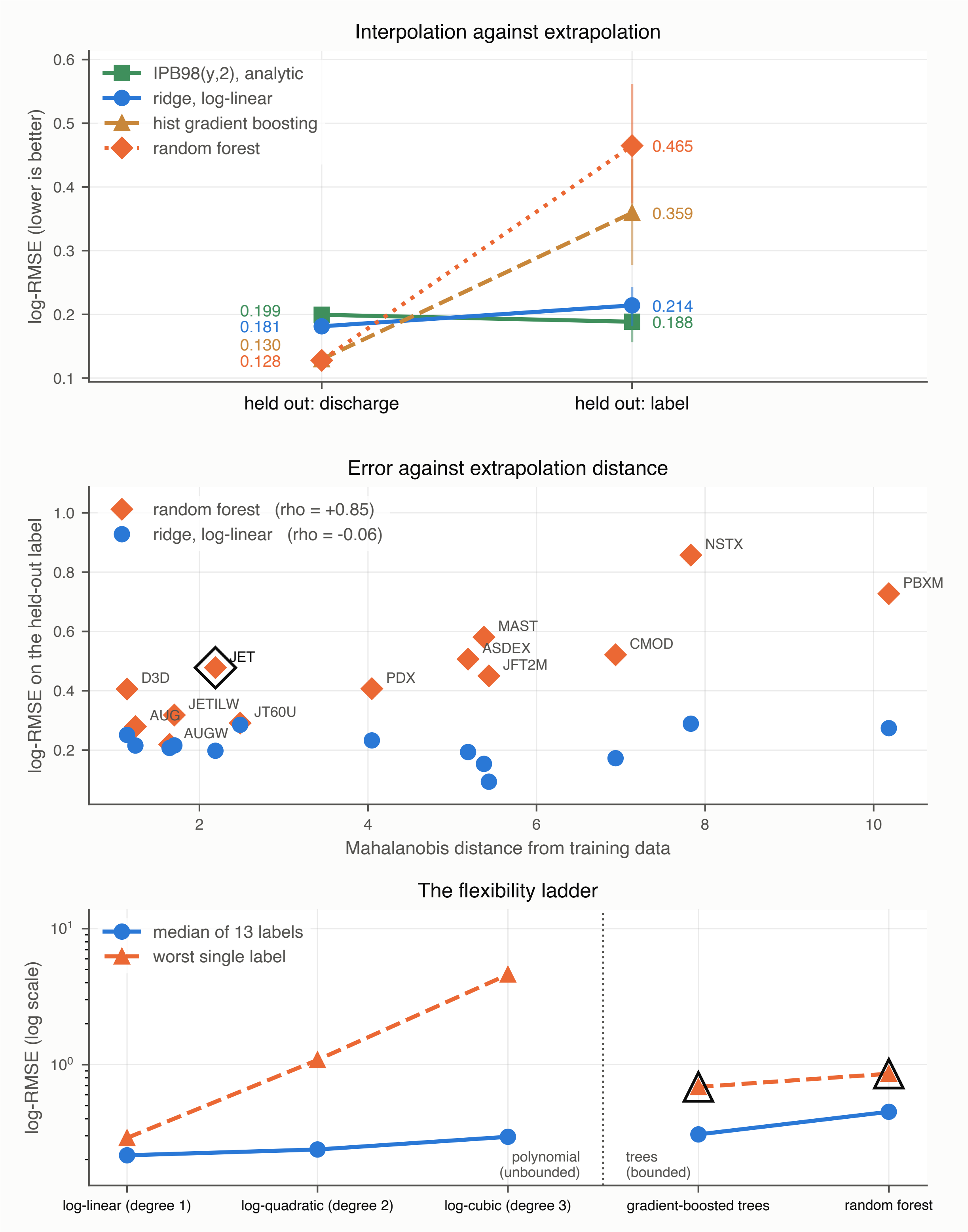

Figure 1: Interpolation against extrapolation. Top: each model under both splits, with bars giving the 95% interval over the 13 held-out labels; the lines crossing is the result. Middle: per-label error against Mahalanobis distance from the training data, rising strongly for the random forest, more weakly for the booster, and flat for the power law. The circled label is the second failure mode, where the true confinement times run above anything the random forest can output. Bottom: the flexibility ladder, where what grows with flexibility is the tail rather than the centre; the circled points are the two tree ensembles, both observed here to stay inside the training-target range, though only the random forest is provably bounded to it.

($p = 0.25$). Neither figure clears a conventional threshold, and the booster's correlation is read throughout as suggestive rather than established. Dropping each label in turn moves the forest's ρ only within $[+0.81, +0.88]$, so no single extreme device carries it. One control changes how the rest should be read: the *mean baseline*, which fits nothing, also correlates with distance, at +0.56. Distance predicts how hard a unit is in general. What is unusual is not that the trees degrade with it but that the power law does not.

Mahalanobis distance between mean log-feature vectors is one measure of how far a unit sits from the training set, and it is the only one used here. It compares first moments through a single pooled covariance, so it is insensitive to differences in the shape or spread of a unit's operating envelope at a fixed mean, and the covariance it uses is singular by Sec. 3 and so is inverted through the pseudo-inverse. Every statement below about "distance" is a statement about that one diagnostic, not about distributional difference in general.

The per-label scores behind that correlation are in Sec. S8 of the Supplementary Material. On the four most distant labels (PBX-M, NSTX, C-Mod, JFT-2M, all small, one of them spherical) the forest's error is 2.7 to 4.8× the power law's; on the four nearest it is within 1.7×.

A hard bound. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time anywhere in the training fold, and its largest held-out target is 3.7× above anything any tree in the forest can output. More rows would close this only if they carried larger targets: within the same target range no amount of additional data, and no tuning, moves a ceiling that is a property of the functional form.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the argument above does not transfer to it. What we can report is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 5, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical label: median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1's worst label is 0.289; degree 2's is 1.083 and degree 3's is 4.601, both on C-Mod. A bounded range is simultaneously why neither ensemble extrapolates and why neither blows up: their worst cases, 0.686 and 0.857, sit far below degree 3's.

Read together, these three diagnostics invite a summary this paper does not endorse. Every flexible model scored so far is also either bounded, like the two ensembles, or divergent, like the cubic, so nothing above can tell apart "flexible models fail out of distribution" from "models that stop trending outside their data fail out of distribution". The two make identical predictions about everything in Table 1. Section 12 builds the model that separates them and supports the second interpretation, so the reader should carry the flexibility reading as provisional from here rather than as the paper's conclusion.

Table 2: The random forest and the power law under both row populations, both aggregations, and both definitions of a machine. The inversion is the pattern of the random forest leading under cross-validation and trailing the power law under leave-one-out; it holds in all eight combinations.

	pooled over rows		each unit equally	
Split	forest	ridge	forest	ridge
CV, all 18 labels	0.128	0.181	0.149	0.238
CV, the 13 scored labels	0.123	0.177	0.124	0.187
leave-one-label-out (13)	0.410	0.214	0.465	0.214
leave-one-device-out (11)	0.551	0.200	0.527	0.212

4.2 Sensitivity to population, aggregation, and device definition

Table 1 attributes a difference to the split, so anything else that differs between its two columns is a confound. Three things do. Grouped CV scores all 6228 rows while leave-one-out scores only the 13 labels large enough to hold out. Grouped CV pools out-of-fold predictions into one log-RMSE over every row, where JET and AUG dominate, while leave-one-out averages per-label scores, where every label counts once. And JET and JET-ILW are one physical tokamak either side of a wall change, as are AUG and AUGW, so holding out JET-ILW while training on JET does not hold out an unseen device.

The labels too small to score. A fourth choice is worth stating because it looks like a confound and is not. Five labels hold fewer than 30 rows (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them, under 1% of the database) and none can carry a held-out score of its own. They are nonetheless in every training fold, since the threshold governs scoring rather than training. They are also among the most unusual points in the feature space, so whether a flexible model can exploit those few rows is a fair question to ask of the ranking.

Dropping them from training as well answers it. Every fitted model gets slightly worse without them, by 0.012 for the power law, 0.027 for the gradient booster and 0.007 for the random forest, and the ordering over the 13 held-out labels is unchanged: power law 0.214 to 0.226, gradient booster 0.359 to 0.387, random forest 0.465 to 0.472. The 43 rows help the tree ensembles slightly, as one would expect of rows in a sparse region, and nowhere near enough to recover the inversion. The analytic reference is unaffected, since it fits nothing.

Table 2 varies all three. The forest leads the power law under every cross-validation cell and trails it under every leave-one-out cell, so the inversion is a property of the split rather than of the population or the aggregation. Collapsing the variants makes the forest's failure worse rather than better: over 11 physical devices its mean gap over the power law grows from +0.251 to +0.315, with a paired 95% interval of $[+0.221, +0.408]$ against $[+0.157, +0.342]$ over the labels, and it loses on 11 of 11. The published law's score is almost unmoved by any of it, 0.188 to 0.184 per unit.

A newer published law. IPB98(y,2) is the ITER reference but not the most recent scaling fitted to this database family. ITPA20 and ITPA20-IL [2] were derived from DB5.2.3 and weaken the size dependence considerably, so they are the obvious second reference. Two things have to be said before any number, because they bound what the comparison can mean. Each law was fitted to a different selection out of DB5.2.3, and each specifies an elongation this database does not supply under that definition; the delivered file carries the areal elongation and nothing that can be converted to the other without the geometry the deposit omits. No cleaning of these rows can therefore score ITPA20 on the quantity it was fitted to, and *no absolute level reported here is a fair test of either law.* None is offered as one.

What survives that is the relative comparison this paper is about, which does not depend on either law being scored on its own terms. Evaluated analytically on the rows analysed here, ITPA20 scores 0.265 over all rows, 0.267 per label and **0.272** across the ITER-size-matched cut; ITPA20-IL scores 0.393, 0.446 and 0.363. At that cut ITPA20 sits against the random forest's 0.938 and the gradient booster's 1.072, factors of 3.4 and 3.9, and essentially level with the fold-fitted power law's 0.278. Despite the definition mismatches, both published power-law forms score substantially below the saturating tree models at that cut. Like IPB98(y,2), neither is refitted within these folds, and ITPA20 and ITPA20-IL were fitted to selections out of the very revision analysed here.

Where the row threshold is set. The 13-of-13 count is reported at a 30-row eligibility threshold, which is a choice, so it is swept. Section S8 of the Supplementary Material scores leave-one-label-out at 10, 20, 30 and 50 rows. Only which labels are *scored* changes: every label stays in every training fold at every threshold, so the models being compared do not change. The power law wins every scored label at 20, 30 and 50 rows, and 14 of 15 at 10 rows, where the exception is TCV. TCV carries 11 rows, which is why it sits below the threshold in the first place, and a per-label log-RMSE over 11 rows is not a quantity this paper would report on its own. The mean gap moves between +0.211 and +0.251 across the four thresholds and the sign test stays below 10^{-3} throughout, so the headline does not depend on where the line is drawn.

One partition, or any partition. The cross-validation column comes from a single deterministic `GroupKFold` assignment of discharges to five folds. Since this paper is about validation protocol, whether the 29% margin is peculiar to that one partition is a fair question. Repeating the comparison over ten shuffled assignments of the same discharges into the same number of folds gives a mean forest advantage of 27.8% over the power law, ranging from 27.0% to 28.5%, with the forest ahead in all ten. The deterministic partition's 29.5% sits just above that range, so the reported margin is if anything mildly favourable to the forest, and the interpolation win it later gives up is a property of the comparison rather than of one fold assignment.

Tuning the flexible models. Every score above uses the prespecified, untuned settings of Sec. 2, which leaves open whether the ensembles lost only because they were not hyperparameter-tuned. We tested that by giving both a grid search nested inside each training fold: three grouped inner folds by discharge over the training rows only, so the held-out discharge fold, label or size cut never takes part in choosing a hyperparameter. The forest searched `max_features` over $\{1.0, 0.5, \mathrm{sqrt}\}$ and `min_samples_leaf` over $\{1, 5\}$, the booster `learning_rate` over $\{0.05, 0.1\}$ and `max_leaf_nodes` over $\{15, 31\}$.

The search does real work on the forest and changes nothing about the conclusion. It moved away from the baseline `max_features` = 1.0, `min_samples_leaf` = 1 setting in 13 of the 19 folds, and it helps where the paper says a flexible model should help: cross-validation improves from 0.128 to 0.126 and the leave-one-label-out mean from 0.465 to 0.403. At the ITER-size-matched cut it makes the forest *worse*, 0.938 to 1.121, against the power law's 0.278. The booster's search returned scikit-learn's defaults in all 19 folds, so its numbers are unchanged.

The inner folds above group by discharge, which is how a practitioner would ordinarily tune. Grouping them by whole machine instead matches the way the model is deployed, and it helps both models: the forest falls to 0.376 and the booster to 0.350. Both remain well above the power law's 0.214, so the reversal survives either selection procedure.

Tuning therefore widens the interpolation margin that later inverts and does not repair the extrapolation failure, which is what the training-target bound predicts: no setting of these hyperparameters moves a ceiling that comes from averaging training targets.

Two conventions this paper does not follow. Neither the population nor the weighting used above matches how the ITPA scalings were produced, and both are choices rather than findings, so both are measured.

IPB98(y,2) was fitted on a selection that excluded spherical tokamaks; this database does not, and MAST and NSTX are two of the 13 scored labels and two of the four most distant. Sec. 5 controls for this only at the size cut. Dropping every label whose median inverse aspect ratio exceeds 0.5, which removes MAST, NSTX and START and 232 rows, from training and scoring alike, leaves the reversal intact: the forest is worse than the power law on all 11 remaining labels, mean gap +0.182, exact $p = 9.8 \times 10^{-4}$, at 0.395 against the power law's 0.212. The inversion is not an artifact of including the spherical devices.

The weighting arm is the one that qualifies the headline. Every fit above weights rows equally, so JET and ASDEX Upgrade, which supply 77% of them, set most of what each model learns. Refitting every model with each label weighted equally hurts the power law more than the forest, 0.214 to 0.259 against 0.465 to 0.447, and the win count falls from 13 of 13 to **12 of 13** ($p = 3.4 \times 10^{-3}$, mean gap +0.188). The label that changes hands is AUG-W, whose unweighted margin was the narrowest of the thirteen at +0.012 and which moves to −0.067. The direction and its significance survive the reweighting; unanimity does not, and the 13 of 13 quoted throughout is the unweighted count.

The win counts support an exact test, under a null that is worth naming precisely. Call it the independent-unit-sign null: each unit, label or device as the case may be, is assigned to either model independently with equal probability. Under it, losing 13 of 13 has two-sided $p = 2.4 \times 10^{-4}$ and losing 11 of 11 has $p = 9.8 \times 10^{-4}$. The device-collapsed count is the one to quote, since the 13 database labels are not 13 independent devices, and the same non-independence is a reason to prefer these counts to the 13-label bootstrap.

That null is an idealisation, and Sec. 2 says in what way: any two of these fits share all but two of their training machines, so the outcomes are correlated through the training set even where the held-out devices are distinct. The descriptive statement carries the claim on its own. The forest is worse on every physical device scored, and the paired interval on the per-device difference excludes zero.

5 A size-extrapolation stress test matched to ITER

Holding out JET still leaves the rest of the database spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER's major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest database labels (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest labels (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of

Table 3: Three questions of increasing difficulty, scored in log-RMSE. The *size cut* column is the ITER-size-matched cut. Reading down it rather than across the table is the point: the two models that win the first column move far toward the constant baseline in the third, landing closer to it than to the power law in absolute error.

Model	discharge CV	LOLO	size cut
IPB98(y,2)	0.199	0.188	**0.194**
ridge, log-linear	0.181	0.214	**0.278**
random forest	0.128	0.465	**0.938**
hist grad. boosting	0.130	0.359	**1.072**
mean baseline	0.869	0.994	1.459

1.823 against ITER's 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

Two things bound what this cut can carry, and they are why it is a supporting stress test here rather than the result the paper rests on. Its held-out side is 96% JET: of 2730 rows, 1762 are JET and 866 JET-ILW, with JT-60U contributing 100 and TFTR 2. So it is close to a single-device holdout at a size ratio, not a survey of large machines, and the per-label scores below matter more than the pooled one because of it. And the boundary is drawn on major radius alone. Hall *et al.* [15] argue that the weak size dependence in this database comes from a subset localised in dimensionless space rather than from size as such, which is a different place to cut; a reader persuaded by that should read this section as measuring transfer across one particular boundary rather than across size in general. The claim the paper rests on is the leave-one-device-out result of Sec. 4, which holds out 11 physical devices one at a time and does not depend on where a size boundary is drawn.

Table 3 places that cut beside the two easier splits. At the cut the power law scores 0.278 against the analytic law's 0.194 and a constant predictor's 1.459. The random forest scores 0.938 and the gradient booster 1.072, so both sit nearer the constant than the power law in absolute error. Asked the question a scaling law exists to answer, the two tree ensembles that lead the conventional cross-validation comparison land closer to a constant than to the published law they beat by 36% under cross-validation. The ordering is identical on all three individually scored held-out labels, so the pooled number is not an artifact of how JET is weighted within the pool.

That control is weaker than it looks, and the composition of the held-out side should be read before the number is. The four labels above the cut are JET (1762 rows), JET-ILW (866), JT-60U (100) and TFTR (2); TFTR falls far below the 30 rows required to report a label on its own and enters only the pooled number. So 2628 of the 2730 held-out rows, 96 % of them, come from a single physical device, and the three individually scored labels are two physical machines rather than three, since JET and JET-ILW are the same tokamak in two wall eras. Seeing the same ordering twice on one machine is not two independent confirmations of it. What this cut establishes is that the tree ensembles collapse across an ITER-sized jump on the rows this database has above one, and what would settle it is a second large device at comparable weight, which this database does not contain. That is a limitation of the question as much as of the split: large machines are few, and their scarcity is what makes the extrapolation hard. The leave-one-label-out result of Sec. 4, which holds on all 13 labels and all 11 devices, is the one that rests on many machines; this one rests on a large jump. Dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is not driven by including the spherical devices; other shape or regime covariates that track size are not ruled out by this control.

6 A power law with a bounded correction

The diagnosis in Sec. 4.1 suggests a specific repair. Fit the power law, learn a correction on its log residuals, and damp it by λ:

$$\log\tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}}. \tag{2}$$

At $\lambda = 0$ this is exactly the ridge row of Table 1. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [8], and this section makes no claim on that idea. What is varied is the one property the diagnosis points at, the long-range behaviour of the correction: one corrector is a degree-2 log polynomial, which is unbounded, and the other is depth-2 boosted trees, which never left the residual range they were trained on here. The two families differ in more than that, and the controlled version of the comparison is Sec. 12.1. The damping factor is swept over nine values from 0 to 1, both correctors carry deliberately strong penalties so that λ rather than their own regularisation does the damping, neither corrector's hyperparameters are tuned, and the whole estimator is refitted from scratch inside every training fold; the exact specification is in `analysis_hybrid.py` in the archived code.

Across the ITER-size-matched cut the two correctors go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out labels and by the widest margin on the most distant of them, JT-60U (0.440 to 0.281).

The mechanism is measurable. Fitted on the 14 small labels, the base power law is *biased* on the held-out ones, at a mean log residual of -0.218, so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves its training range on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same bounded range that makes these ensembles useless as predictors of τ on a larger machine limits how much extrapolation error the correction can contribute, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero. Bounded is not the same as right: a correction confined to that range can still point the wrong way, and what is measured here is that on these rows it does not.

Two limits qualify every later summary of this section. Cross-validation selects $\lambda = 1$ in both families while the rung best on a held-out label is $\lambda = 0$, so a practitioner tuning on the split they would ordinarily use does not arrive at the rung that transfers. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, the bounded hybrid equals the power law at $\lambda = 0$, where it is that model, and beats it at all eight non-zero damping settings swept there, with the mechanism measurable.

7 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. We calibrate split-conformal intervals [16] on log residuals, holding back 25% of *discharges* to calibrate.

Table 4: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOLO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	**0%**	1.23×
random forest	91%	35%	**3%**	1.23×

The construction is specified as follows, so that a reader can reproduce the coverage numbers rather than infer the procedure from them. Within each training fold, the distinct discharge identifiers are permuted with `numpy.random.default_rng` seeded at 20240811, and the first round$(0.25\, n_{\text{discharges}})$ of the permutation are the calibration discharges; every row of a calibration discharge is a calibration row, and the model is fitted on the remaining rows only. Arms with fewer than 100 calibration rows are not reported. The conformity score is the absolute log residual $|\log \hat{\tau} - \log \tau|$. The half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score in ascending order, at $\alpha = 0.10$, which is the finite-sample conformal rank rather than the plain empirical quantile; the $+1$ accounts for the test point itself being one of the $n+1$ exchangeable scores, and where that rank exceeds n the half-width is reported as infinite rather than as a finite but invalid number. Ties need no rule, since the quantile is a rank on a sorted array. The interval is symmetric in $\log \tau$ and is a pointwise interval for a new observation, not for a latent mean. Under grouped cross-validation the per-fold coverages are pooled over rows; under leave-one-label-out each held-out label is calibrated and scored on its own fold and the coverages are then averaged over machines.

Holding out whole discharges keeps a calibration row and a test row from coming out of the same shot, but the conformity unit is still the row, and rows arrive in correlated clusters of quasi-stationary slices from one discharge. The standard finite-sample guarantee assumes exchangeable conformity scores and clustered rows do not supply that, so every coverage number below is empirical row-level coverage rather than a guaranteed rate. A discharge-level conformity score, or a cluster-aware conformal method, would be the way to recover a guarantee, and neither is used here.

That proviso is what the section measures. Empirical coverage is near nominal under grouped CV, where every model lands within a point of 90%, and that control is what licenses reading the rest of Table 4. It is a useful in-distribution control rather than a verified guarantee: clustered row-level conformity scores do not satisfy the finite-sample exchangeability condition either, so what the CV column shows is that the construction behaves as intended when the test rows come from labels the model has seen. The exchangeability assumption is no longer justified when the test rows come from a systematically held-out label absent from the model's training and calibration distribution, and what that costs empirically is measured below. That exchangeability is the binding assumption, and that repairing it requires knowing the shift, is the subject of the weighted-conformal literature [17], and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one [18]. The random forest's 90% interval then covers 35% of rows on an unseen label and 3% across the ITER-size-matched cut; the gradient booster covers none of the 2730 rows at all.

The widths barely move: no model's interval changes width by more than 1.5% between the two arms, because the half-width is set by calibration rows drawn the same way in both.

The intervals do not become vague out of distribution. They stay the same size and miss. Figure 2 shows both halves of that: the coverage of every model under each split against the nominal line, and the per-label coverage against extrapolation distance.

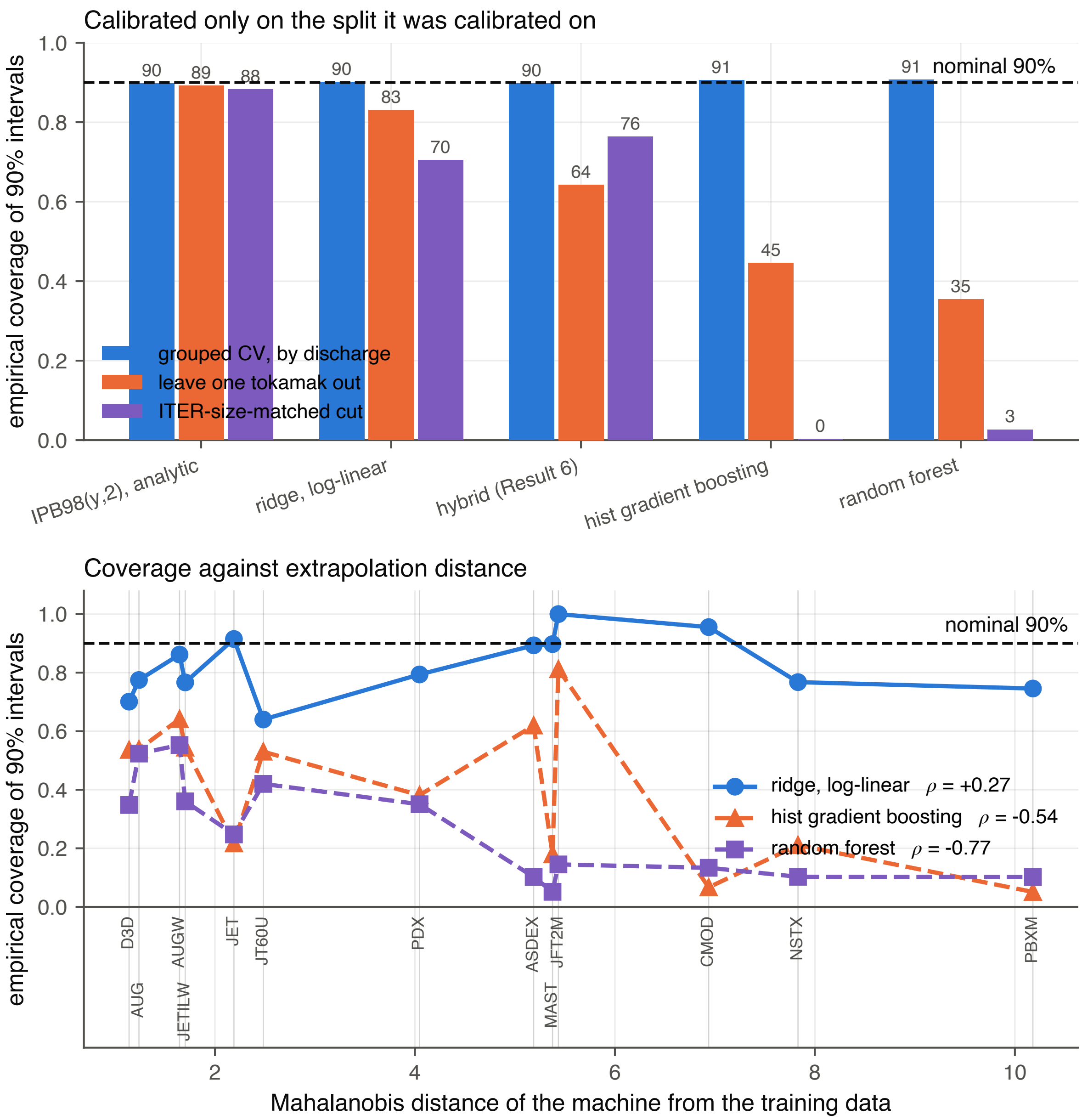

Figure 2: The interval collapse. Top: empirical coverage of nominal 90% split-conformal intervals for each model under the three splits, against the nominal line. Every model is calibrated under grouped cross-validation, and only the linear and constrained fits stay near it once the held-out unit becomes a whole label and then a size range. Bottom: per-label coverage against the Mahalanobis distance of the label from the training data. Under grouped cross-validation coverage is flat on the nominal line at every distance; under leave-one-label-out the tree ensembles fall away with distance and the power law does not. Interval widths behind both panels move by under 1.5% between the splits, so what is drawn is intervals that keep their size and lose their coverage.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the per-label error correlations of Sec. 4.1 with the sign flipped. The power law's intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction. It is a measurement of an assumption being false.

8 Dimensional constraints from Connor–Taylor similarity

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [19] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents.* A physics assumption is therefore a matrix C, and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \tag{3}$$

which the Karush–Kuhn–Tucker (KKT) system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 3.1.

We derive the constraints in code from the definitions of ρ^*, β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. Written as exponents on (length, field, temperature, density),

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \qquad \beta \sim n\,T\,B^{-2}, \qquad \nu^* \sim n\,L\,T^{-2}, \tag{4}$$

and a law expressible in whichever of these a model holds fixed must be invariant under every scale transformation that leaves those groups unchanged. Each such transformation contributes one linear equation in the exponents, so the admissible set is a surface and the number of constraints is set by how many groups are fixed:

constraint	groups held fixed	rows of C
Kadomtsev [20]	ρ^*, β, ν^*	1
collisionless	ρ^*, β	2
electrostatic	ρ^*	3

Fixing fewer groups admits more transformations and therefore imposes more constraints, which is why the hierarchy runs the way it does. C and d are built by `dimensional.constraint_matrix` from Eq. (4) alone, and their numerical rows are in `results/dimensional.json`.

Constraining a confinement scaling this way is not new, and IPB98(y,2) is itself an instance of it: the high-β Kadomtsev constraint was enforced when it was fitted, alongside a magnetic-field exponent fixed at 0.15 [1, 2]. Its exponents accordingly land on the Kadomtsev surface [20] derived here at a distance of 0.00096, and on the collisionless surface at 0.0045. The first of those is expected rather than surprising, and we use it only as a check that the surfaces derived in code from Eq. (4) agree with the constraint the published law was fitted under. What is new here is not the idea of constraining the exponents but the comparison: a hierarchy of such surfaces, fitted and scored under a validation protocol matched to the deployment problem.

Table 5: Fitting under the Connor–Taylor constraint hierarchy, ordered by score at the ITER-size-matched cut. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter. The Kadomtsev row is the prospective one, fixed by the literature before this analysis; the collisionless row is the best of the three and was selected after these scores were seen.

Model	in sample	CV	LOLO	ITER cut
power law, collisionless	0.1818	0.182	0.206	**0.183**
IPB98(y,2), analytic*	n/a	0.199	0.188	0.194
bounded hybrid (Sec. 6)	n/a	0.151	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.254
power law, unconstrained	0.1808	0.181	0.214	0.279

*historical reference, fitted to the DB2.8 predecessor of this database [2] and not refitted within these folds. The unconstrained row is the closed-form least-squares fit this section's solver produces, at 0.2790 across the size cut; the ridge spelling of the same power law reported elsewhere gives 0.2778, and Sec. 3 shows the choice between them carries nothing.

The three rungs do not carry the same evidential status, and separating them matters more than ranking them does. The Kadomtsev rung is *prospective*. It was specified by Kadomtsev in 1975 [20], and enforced when IPB98(y,2) was fitted in 1998 [2], both without reference to anything measured here; imposing it is a prediction this analysis inherited rather than a surface it selected. The collisionless rung is the best of the three at the size cut and was chosen after those scores were seen. The prospective one comes first below.

Take the prospective rung first. The Kadomtsev constraint is *free*: at 0.1808 in sample it matches the unconstrained fit to four decimals, because the data already satisfies it unaided. It nonetheless beats that unconstrained fit at every one of the 15 cuts in the size sweep, including all 8 well-powered ones, where a cut is called well-powered when it trains on at least 1000 rows. Satisfying a constraint on average is not the same as being unable to violate it, and what the constraint buys is a fit that cannot wander along that direction when the training half is small or shifted. The underpowered rungs show it directly: at six training machines the unconstrained fit scores 0.971 and the Kadomtsev fit 0.274. Nothing in that comparison was selected after the fact, which is why it is the result in this section to weigh most heavily.

The collisionless rung is the stronger assumption and the better score, and it was chosen post hoc. Something can still be said for it that does not depend on the score. It is a physics judgement about the machine being extrapolated *to* rather than a hyperparameter: one adopts it because one expects the target plasma to be collisionless, and ITER is expected to sit at lower normalised collisionality than the devices in this database rather than higher. That points at this rung in advance, for a reason external to the data. The rung also has no within-surface quantity to tune, so there is nothing for a validation split to fit even in principle. Neither observation makes the result prospective, and the disclosure stands as written.

At the ITER-size-matched cut the collisionless power law scores **0.183**, the lowest held-out error of any model fitted here without using the held-out rows (Table 5). The fit is blind to the held-out rows; the *search* is not. No coefficient ever saw a held-out row, but the constraint rung was chosen after seeing how each of the three scored at this cut, so the cut took part in model selection even though it never entered a fit. This is therefore the lowest error among the surfaces examined, not a prospective demonstration that the collisionless surface is the right one. It beats the bounded hybrid of Sec. 6, which was built specifically for that cut, and it beats the analytic law, which was fitted under a Kadomtsev constraint of its own on a predecessor database. Figure 3 shows the trade between what each constraint costs in sample and what it buys at the size cut.

The collisionless constraint costs 0.001 in sample against the unconstrained fit and wins at 8 of 8 well-powered cuts, so it is not a single lucky cut, but it is worse than Kadomtsev at every rung below the well-powered line: it is the stronger assumption and it pays only where there is enough data for constraining to be worth doing. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it. Every constrained fit scored slightly worse than the unconstrained one under the cross-validation used here, so CV supplies no signal pointing at the constraint that transfers best. That is an observation about these folds rather than a theorem: a hard equality constraint can only match or worsen the *training* residual, and nothing forces it to worsen a held-out score.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 3.1, is worth much less. That comparison sweeps three shrinkage geometries (isotropic, spectral and truncated) over the same nine penalty strengths $10^{-3}, 10^{-2}, \ldots, 10^{5}$, tabulated in full in `results/dimensional_prior_sweep.csv`. Targeting the shrinkage beats isotropic shrinkage at every one of those matched penalty strengths, which is a real effect, but nothing in that family beats simply adopting IPB98's published exponents outright. The hard constraint used here restricts the fit to a surface the Connor–Taylor similarity assumptions permit, where the Gaussian prior tested here favours a location and a direction in parameter space, and in this experiment the surface transfers better. That is a statement about these two families rather than about priors in general: a prior can favour a subspace, and approaches a hard constraint as its variance goes to zero.

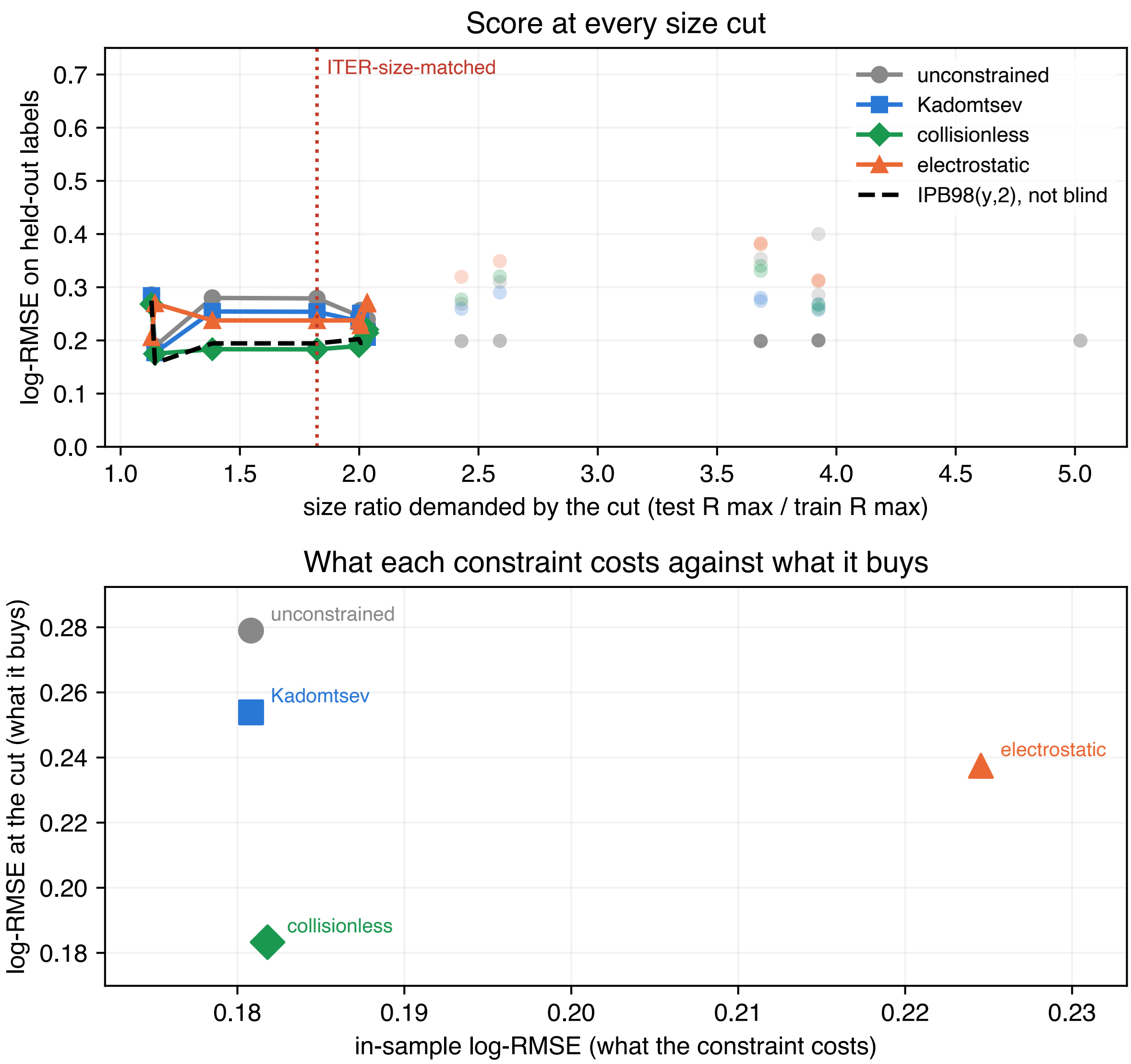

Figure 3: Cost against benefit across the size sweep. Top: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts drawn faint and excluded from claims. Bottom: what each constraint costs in sample against what it buys at the ITER-size-matched cut. The constraint costs almost nothing where the data is plentiful and pays where it is not, and more physics is not monotonically better: the hierarchy has an optimum in its middle, at the collisionless rung. That rung was selected after these size-cut scores had been seen, so it is the best of the three surfaces examined rather than a prospective prediction.

9 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 7 makes a testable prediction. If the collapse is about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves hold. On a held-out label, empirical row-level coverage returns to within two points of nominal for every model, the random forest going from 35% to 88%. Across the ITER-size-matched cut none of the schemes tested restores it, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap, 3% to 40% for the forest. A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

Repeating the whole procedure over the 11 physical devices, which removes the overlap that leaves JET in the calibration set when JET-ILW is the target, shows the repair is less complete than the label-level numbers suggest: the booster reaches 77% by device against 90% by label, and the trees buy their coverage at 2.3 to 2.8× multiplicative half-width where the linear models stay near 1.4×. Among the models refitted within folds, the constrained power law of Sec. 8 is the only one whose plain split-conformal coverage stays near nominal at the size cut, at 91%, and its intervals hold there under every scheme.

Section S1 of the Supplementary Material gives both calibration schemes in full, the two coverage tables, and the reasons neither scheme is a finite-sample-valid group-conformal procedure.

10 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis. The same OSF project publishes the full DB5.2.3 revision, 14153 rows against STD5's 6228, pinned by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it, and leaves two populations this study had not analysed: 5358 H-mode rows STD5 does not contain, across 12 labels with zero row overlap, and 3860 ohmic and L-mode rows in a different confinement regime, which we score against ITER89-P [21] because an H-mode law is the wrong baseline for an L-mode plasma.

The ranking inverts in both arms, and it survives dropping the 336 discharges the disjoint H-mode arm shares with STD5 at different time slices. Counting label–model pairs where a tree ensemble loses to the published law on an unseen label gives 19 of 24 and 10 of 10. Section S2 of the Supplementary Material reports both arms in full, with the positive and negative checks on the row matcher, whose silent failure would otherwise reproduce Sec. 4 by construction. This is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

11 A locked forecast, and a direction that is not size

Everything above is retrospective. To make the disagreement checkable rather than merely described, what each model predicts for three real devices, SPARC, JT-60SA and ITER, is recorded with intervals, a lock date of 31 August 2026 and a content digest over the input rows, so that a later edit leaves a mark. The parameter sets are published design values, and none of the three has a *measured* H-mode thermal confinement time at the operating point tabulated, which is what makes these predictions rather than retrodictions. Section S3 of the Supplementary Material gives the table, the interval construction, the disclosures attaching to it and the falsification conditions; the locked values are in `results/forecast.json`. It is a record rather than a validation, and only JT-60SA is checkable in the near term.

On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**, and for the random forest that gap cannot be closed by tuning: its prediction cannot exceed the largest training target, 1.321 s, where the linear and constrained models predict 2.84 s to 3.59 s.

One feature of the exercise was not anticipated by anything above. SPARC sits *further* from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine: its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

12 Flexibility, or long-range saturation?

Section 4.1 left two readings open, and every model scored to this point confounds them. Flexibility is how much structure a model can learn beyond a power law; *saturation* is whether its predictions stop tracking the features once the input leaves the training support, either because they cannot leave a fixed range or because they revert to a fixed value. The forest is the strict case, provably confined to $[\min y_{\text{train}}, \max y_{\text{train}}]$; the booster never left that range on any split scored here; a squared-exponential Gaussian process reverts to its prior mean. What the three share is the absence of a trend that continues outside the data. The ensembles are flexible and saturating, ridge is neither, and the degree-3 polynomial is flexible and non-saturating but divergent, which is why its worst label is 4.601. Nothing scored so far is flexible, non-saturating and well behaved at long range at once.

A Gaussian process (GP) [22] supplies one, and the property is a choice of kernel rather than a tuning parameter. Section S6 of the Supplementary Material fits three over the same log features, with the same optimiser on the same rows, differing in whether the kernel admits a trend that continues at long range, and the ordering is the one saturation predicts. The squared-exponential rung alone scores 1.948 at the ITER-size-matched cut, worse than a constant predictor, its error tracking distance at $\rho = +0.65$. Adding a dot-product term gives the best cross-validated score in this work, 0.112 against the forest's 0.128, together with 0.191 at that cut and an error flat against distance at $\rho = -0.01$. It is also the only flexible model tested whose *native* posterior intervals reach nominal coverage out of distribution without conformal recalibration, covering 92.5% of held-out rows at nominal 90% where split-conformal intervals give the forest 3% and the booster none. The constrained power law of Sec. 8 holds near nominal there too, at 91%, but through an externally calibrated interval rather than one the model supplies itself.

That ladder suggests the mechanism without isolating it, because a dot-product term enlarges the function class as well as changing the asymptote. The rest of this section removes that objection.

$m(x)$ in Eq. (5)	CV	held-out	ITER cut
constant	0.142	0.541	1.948
fitted power law	**0.115**	**0.212**	**0.278**

Table 6: One RBF covariance family and fitting procedure, two mean functions. The constant-mean row reproduces the squared-exponential rung of Sec. S6 of the Supplementary Material to six decimals, which is the control: the decomposition is the same model rewritten. Changing only the mean-function specification moves the ITER-size-matched cut by a factor of 7.0.

12.1 The same experiment with the nonlinear component specified identically

Write the model as a mean function plus a residual process,

$$\log \tau = m(x) + g_{\mathrm{RBF}}(x), \tag{5}$$

and specify g identically in both arms: same RBF covariance family, same tuning subsample, same seed, same optimiser. Only m changes, from a constant to a fitted log-linear power law. The two fitted processes are not the same function, because each is conditioned on its own mean's residuals and each re-optimises its hyperparameters on those. What is held fixed is the nonlinear capacity available to the model and the procedure that fits it, so no part of the difference below can come from one arm having more of either.

Table 6 reports the two arms. Take the control first: with a constant mean, Eq. (5) returns 0.141533, 0.540967 and 1.947535, which is the RBF row of that rung to every digit computed. Nothing has changed except how the model is written. Giving the same process a power-law mean then moves the ITER-size-matched cut from 1.948 to 0.278, a factor of 7.0, and improves the cross-validated score as well, 0.115 against 0.142. Because the residual GP has the same covariance family, the same capacity and the same fitting procedure in both rows, the difference cannot be attributed to adding nonlinear model capacity.

The second row's cut score is worth reading closely: 0.277823 is plain ridge's score at that cut, to six decimals. Far outside the data the RBF residual has reverted to zero, which is what an RBF process does, and the model falls back to its power-law mean, which is the mechanism of this section stated as a single number. A saturating model reverts to something; the failure of the squared-exponential rung is that a constant mean makes that something a constant, and the repair is not to make the model less flexible but to give it something worth reverting to.

Across the model families tested here, this resolves the choice Sec. 4.1 left open: extrapolation failure tracks long-range saturation rather than flexibility alone. The set of models in Sec. 4 is measured correctly, and every flexible member of it was either bounded or divergent, so nothing in it could have separated the two properties; that is why the reading there was flagged as provisional rather than asserted. "Do not use a flexible model out of distribution" is the wrong lesson. "Do not use a model whose predictions stop tracking the features once the input leaves the training data" is the right one, and the random forest's training-target ceiling is its strictest case. This does not displace the constrained power law, which still wins the ITER-size-matched cut with no within-surface hyperparameter to tune, and the process does not beat the power law on a held-out label (0.218 against 0.214).

A Gaussian process also supplies an interval by construction rather than by calibration, so Sec. 7's question can be put to a model that was never calibrated. At the ITER-size-matched cut, at nominal 90%, the flexible unbounded rung covers 92.5%, where split-conformal intervals give the random forest 3% and the gradient booster none. It is the only flexible model tested whose *native* posterior intervals reach nominal coverage out of distribution without conformal recalibration. That is a statement about where the interval comes from rather than about coverage alone: the constrained power law of Sec. 8 also holds near nominal at this cut (91%, Sec. 9), but through an externally calibrated split-conformal interval rather than one the model supplies itself. The bounded rung is the instructive failure: it is the one model that does become vague out of distribution, with a half-width four times any other, and it still covers 16.7%, having reverted to a prior mean nowhere near the answer. Width was never the problem.

13 Discussion

The practical consequence of Table 1 is narrow and specific. It is not that machine learning has nothing to offer confinement scaling, and not that the models scored here are badly built. It is that conventional within-database evaluations do not measure the quantity the law is used for. Cross-validation grouped by discharge, or by shot, or by time slice, holds out rows from machines that remain in the training set. It measures interpolation within known devices. A scaling law is used to size a device that does not exist, which is extrapolation to an unknown one, and on this database the two comparisons do not merely differ in magnitude: they order the candidate models in opposite directions. A gain reported under the first is not evidence about the second.

That is stronger than saying the interpolation number is uninformative. The convention selects on the cross-validated score, and on this database that score picks the random forest, which is the worst of the three on a label it has not seen. A practitioner following current practice is therefore not left with an open question about extrapolation. They are led to the model that extrapolates worst of the three, and the more thoroughly they follow the convention the more confidently they are led there.

13.1 What this changes in practice

The consequence is a reporting convention rather than a modelling one, and it is short enough to state as a checklist. For any confinement model offered as an alternative to a power law on this database:

1. Report leave-one-device-out beside cross-validation. It costs one refit per device, on the same rows and the same features, and the two numbers print together. Where they agree the interpolation number is informative. Where they disagree, as here by a factor of 3.6, the disagreement is the result, and the interpolation number should not be quoted on its own.

2. Report how far the target operating point sits from the training distribution. This needs no held-out device. The Mahalanobis distance of a label's mean log-feature vector tracked per-label error at $\rho = +0.85$ for the random forest and at $\rho = -0.06$ for the power law, so it separates the models that will degrade from the ones that will not, without waiting for the machine to exist.

3. Say whether the model's predictions can leave the range of its training targets. A random forest provably cannot, since every prediction is an average of training tar-

gets, and a model that cannot exceed the largest confinement time it has seen cannot project a machine expected to exceed it. That is a property of the estimator, checkable before anything is fitted.

4. Do not carry a calibrated interval across a device boundary without saying so. Split conformal assumes exchangeable conformity scores, and across a held-out device that assumption is false. What it costs is not a wider interval but the same interval with the coverage removed: 90% nominal delivering 35% on an unseen label and 3% across the size cut, at a width that moves by under 1.5%.

Only the first needs a refit. The other three cost nothing and are computable from the fitted model and the intended operating point alone.

The positive findings point the same way, and both are about the far field rather than about capacity. Constraining the exponents to a Connor–Taylor surface and adding a linear component to a Gaussian process kernel are unrelated operations, and both supply long-range structure. They are not equivalent in what else they change: the constraint removes freedom from the fit, while the linear kernel also enlarges the GP's function class, so that rung on its own cannot separate the asymptote from the capacity. The mean-function ablation of Sec. 12.1 is the cleaner test, since there the nonlinear covariance family and the procedure fitting it are held fixed and only the long-range mean changes. It should still be read as a statement about the families tested, not about flexible models in general.

For practice, the cheaper option was also the stronger one out of distribution. A power law refitted on the same rows is a stronger baseline out of distribution than any tree ensemble scored here, and it costs a single least-squares solve; the estimator hardly matters, since ordinary least squares on the independent columns reproduces the reported ridge to within 0.0012 on every split. Where a flexible model is wanted, the evidence here favours one that keeps trending, and the linear-plus-RBF Gaussian process was both the best cross-validated model in this study, at 0.112, and the only flexible model whose native posterior intervals reached nominal coverage out of distribution without conformal recalibration; the constrained power law reaches comparable coverage there through split conformal instead. Where a physics constraint is available, imposing it is close to free and helps out of distribution. The rung the literature fixed in advance costs nothing in sample and beats the unconstrained fit at all 15 cuts of the size sweep; the rung selected after those scores had been seen costs 0.001 and was the best-scoring option at the size cut.

None of this is settled by one database. Running the same audit on two scaling laws outside fusion, reported in Secs. S4 and S5 of the Supplementary Material, reproduces the extrapolation failure in both while the ranking inversion appears only where the flexible model first wins the interpolation split [23, 24]. The practical reading is the weaker one: where a flexible model has no interpolation advantage to lose, the extrapolation hazard is still present, with no warning sign at all.

14 Limitations

The refit population is not IPB98's, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of the 18 database labels are scored under leave-one-label-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen labels is a small bootstrap and the paired differences are the more reliable statistic.

The training-target bound of Sec. 4.1 is a theorem for the random forest and an observation for the gradient booster: boosting sums tree outputs rather than averaging targets, so nothing confines it to the training range, and what is reported here is only that it stayed inside that range on all 14 splits scored. The ITER-size-matched cut reproduces ITER's size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient.

The per-label coverages in Sec. 7 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 8 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one.

The robustness arms of Sec. 10 are disjoint in rows but not independent in provenance: both come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone. The H-mode arm is disjoint in rows and not, as first constructed, in discharges; Sec. 10 reports the rerun with all 336 shared discharges removed, which the reversal survives. The locked forecast of Sec. 11 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires an H-mode campaign rather than the completed L-mode one and we put no date on when that arrives, and it is the one device of the three that sits inside the database's size range.

The two biological replications carry their own limitations, stated in full in Secs. S4 and S5 of the Supplementary Material: the metabolic-rate arm has one predictor and eleven groups that are phylogenetically related rather than independent, which the comparative-biology literature handles with phylogenetically controlled regression [25] and we do not, and the plant ladder is one ordering of one selectively intersected set of predictors, so it establishes that a crossing exists and that the inversion tracks it, not that three is a threshold. Neither establishes a rate for anything.

Every power-law fit here treats the engineering parameters as measured without error, which the confinement-scaling literature has long objected to because these predictors carry real measurement uncertainty. Refitting Eq. (1) by orthogonal distance regression, which spreads the error across the predictors instead, moves the exponents enormously: up to 5.6 on the inverse aspect ratio, with the major radius going from 1.58 to 5.40. It also more than doubles the in-sample error, 0.454 against 0.181. Both are symptoms of one thing rather than a better fit. With no per-variable measurement uncertainties published for this database, an errors-in-variables (EIV) estimator has nothing pinning it down along the weak singular direction of Sec. 3.1, which carries 0.3% of the design's variance, and it runs away along exactly that direction. The honest statement is that these are least-squares exponents, that an EIV fit would differ, and that this database does not supply what computing a credible one would need.

Every fit also weights rows equally, so JET and ASDEX Upgrade set most of what each model learns even though the target is transfer to a device neither resembles. Refitting every model with each label weighted equally, reported in Sec. 4.2, costs the power law more than the forest and takes the win count to 12 of 13. The row imbalance is therefore not the source of the power law's advantage, but it is not irrelevant to its size either, and one weighting is not a characterisation of how the result behaves under reweighting in general.

The Gaussian-process intervals of Sec. 12 are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term and are exponentiated to τ; that is the closest available analogue of a conformal interval, which is also a statement about a new observation rather than a latent mean, but the two are not the same construction and the comparison should be read as such. Those kernels are tuned by marginal likelihood on a seeded subsample of each training fold, with only the hyperparameters coming from the subsample and the posterior then conditioned on the whole fold, which is stable across the range we could afford (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and "linear plus RBF" is one reading of "a power law with a bounded correction" among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 8 all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

15 Conclusion

On real confinement data, the model that wins cross-validation by 29% over a log-linear power law refitted inside the same folds loses to that same power law on all 13 held-out database labels and on all 11 physical devices once the JET and ASDEX Upgrade variants are collapsed, and to the published IPB98(y,2) law as well, though that law is a historical reference fitted to the DB2.8 predecessor of this database and was not refitted within these folds. Three separate diagnostics point at that failure and none of them is a tuning fix; the mechanism they converge on is long-range saturation. At the size extrapolation that separates this database from ITER, the two tree ensembles that lead the conventional comparison land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. The inversion survives rows the standard set does not contain, a second confinement regime, the collapse of 13 labels into 11 devices, either aggregation of the score, a nested hyperparameter search under two selection procedures, and shuffled fold assignments, so it is not an artifact of a single bookkeeping choice. What the models disagree about is recorded rather than argued: on a machine inside the database's size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

The repair that scored best is not additional model flexibility at all: it is a dimensional constraint on the existing power law, and the two rungs tested carry different weight. The Kadomtsev rung was fixed by the literature in 1975 and enforced when IPB98(y,2) was itself fitted, so imposing it here selected nothing; it costs nothing in sample and beats the unconstrained fit at all 15 cuts of the size sweep. The collisionless rung gave the lowest observed error of any fit tested here at the ITER-size-matched cut, but was selected after those scores had been seen, so it is the lowest error among the surfaces examined rather than a prediction that this surface is the right one.

Two results temper the diagnosis and neither weakens the practical advice. In the nested predictor ladder of Sec. S5 the extrapolation failure persists at every predictor count tested while the ranking inversion does not, because the inversion needs the flexible model to win the interpolation split first: a field whose scaling law has one predictor faces the same danger with no warning at all. And separating flexibility from long-range saturation shows the failure belongs to the second. Holding the nonlinear component's specification fixed and varying only the mean function moves the ITER-size-matched cut by a factor of 7.0, and the better arm lands exactly on the power law's score there, because far from the data its correction has reverted to zero and left the trend behind.

The lesson is not that flexible models cannot be trusted beyond their data. It is that a model whose predictions stop tracking the features once the input leaves the training data is structurally unsuited to a problem where the target itself has to leave that range. The two are easy to confuse because the flexible models that win the conventional interpolation comparison here are the saturating ones, while the non-saturating polynomial control becomes unstable in extrapolation instead.

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Use of generative AI. Two generative AI systems were used in preparing this work, and each is named with its version and its role.

Anthropic Claude Opus 5 (model `claude-opus-5`) assisted with developing, debugging and executing author-specified analysis code in the accompanying repository, and with drafting and editing passages of this manuscript.

OpenAI GPT-5.6 Sol was used to critique complete drafts against journal policy and for scientific and historical accuracy, to support the literature review, and to identify methodological issues and propose robustness checks. Several of its suggestions were adopted after the author judged them worth pursuing. They include the corrected account of how IPB98(y,2) was fitted, in Sec. 4 and Sec. 8; the recomputation of the distance correlation over the 11 physical devices in Sec. 4.1; the device-level calibration of Sec. 9; and several of the references in Sec. 3.1 and Sec. 11, which the author then located and checked against their published records.

Neither system is an author. The author designed the study, decided which suggestions to pursue, specified and implemented every analysis reported here, verified every reported number against the generated artifacts, checked every reference against its published record, and takes full responsibility for the content, the conclusions and any error in them. No text was incorporated without the author reading and verifying it.

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Competing interests

The author declares no competing interests.

Author contributions

Sole author. L.S. designed the study, wrote the analysis code, performed the analysis, and wrote the manuscript.

Data availability

Four third-party datasets underlie this work. None is redistributed here; each is fetched from its public source by a script in the repository and verified against a pinned SHA-256 on load, so every number above refers to one specific set of bytes rather than to whatever a host currently serves.

- **ITPA Global H-mode Confinement Database, standard analysis set STD5 (DB5.2.3).** The basis of every fusion result. Openly available from the OSF deposit *International Global H-Mode Confinement Database* [26], described in Ref. [2]. File `hdb5_std5.csv`.
- **The full DB5.2.3 revision.** Used only for the replication of Sec. 10, which scores rows STD5 excludes. It is a separate download from the same OSF project (`https://osf.io/download/zhwa3/`) and is not a superset this paper generated. File `hdb5_db523.csv`.
- **Mammalian basal metabolic rate.** The Figshare deposit of Capellini, Venditti and Barton [27]. Used in the Supplementary Material. File `allometry_bmr.txt`.
- **Biomass And Allometry Database (BAAD) v1.0.1.** Released CC0; the data paper is Ref. [24] and the archive analysed here is release v1.0.1 [28]. Used in the Supplementary Material. File `baad_data.zip`.

The pinned fingerprints are:

file	SHA-256
`hdb5_std5.csv`	`67601c2da5c51f90cf6298ff499cccc7` `4d09ac80c2b98c7dde0d8db3ebb9ac5b`
`hdb5_db523.csv`	`7a48e34379663e3e298924990f05cd8f` `16b8581516bcfa7bb8f438f83ae80ab6`
`allometry_bmr.txt`	`ab22d9ae8f35e96d7fbf5d8e53053d64` `7f0594d74f8917888169993ce668571e`
`baad_data.zip`	`0375f012475658c039e3dcef398951e8` `f68cdeb5430eec8cd65548965496cfe2`

The locked forecast of Sec. 11 additionally carries a content digest over its input rows, so a later edit to the operating points leaves a mark:

```
960d537ee3df8780afd2054e51d01a268
f8803aca64a830dba37fd1ff5319192
```

All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper and its Supplementary Material are openly available under the MIT licence at `https://github.com/lsalcedo100/FusionFlux`, and the exact version used for this paper is archived with a persistent identifier [29]. The analysis code is pinned as firmly as its inputs: every number and figure above was produced at commit

```
a6a15407c592cb7abfa6cf12d80e1e9bbc18a42c
```

Every script verifies that fingerprint against the pin as it loads the data and refuses to run when the two disagree, so no number above can have been computed from a different revision of the database. Several of the JSON outputs under `results/` also record the digest they verified, which carries that provenance into the artifact itself.

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